

# claude-windows / aa329f09-28b7-4acc-b062-98ec7e905abc

## Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\aa329f09-28b7-4acc-b062-98ec7e905abc\subagents\workflows\wf\_ddd29371-8c2\agent-a26061bfa9cf1879b.jsonl`
- SHA-256: `a570252b12196a20592387440c2c869565b6c2f5860b513b995cec1916c0af9e`
- Source modified: `2026-05-30T03:08:00+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf\_ddd29371-8c2`
- session\_id: `aa329f09-28b7-4acc-b062-98ec7e905abc`

## Transcript

### 1. user (2026-05-30T03:03:23.569Z)

#### Web Searcher: ffmpeg GPL/LGPL + video codec patent royalties (AVC/HEVC/AAC) and royalty-free outputs

Research question: "RESEARCH GOAL: Identify and eliminate every dependency that could legally or economically hinder monetizing the "badminton-highlight-indexer" project as a HOSTED API SaaS. Business model: the project runs server-side on a cloud GPU VM (NVIDIA); users upload their own badminton videos and receive auto-generated highlight reels. CRITICAL CONSTRAINT: we do NOT distribute the code or any binary to users ? everything stays server-side ? so software that is merely \*hosted\* on the server (invoked as a separate process, e.g. ffmpeg) is acceptable where distribution-triggered copyleft (GPL/LGPL) would NOT be triggered, BUT network/SaaS copyleft (AGPL §13) and external metered-API terms STILL apply. Preference order: permissively-licensed (Apache-2.0/MIT/BSD) and royalty-free, self-hostable components. Paid commercial licenses (e.g. Ultralytics Enterprise, codec patent licenses) are acceptable ONLY as clearly-flagged fallbacks. Output must be decision-grade with licenses verified against PRIMARY sources (actual GitHub LICENSE files, official pricing/terms pages), because this drives a real business decision ? flag any claim you could not verify.

CURRENT STACK TO AUDIT (verify each license + its SaaS implication):

- ultralytics (Python lib) + the bundled YOLOv8n.pt model weights ? believed AGPL-3.0. Confirm the license of BOTH the code and the pretrained weights, whether the AGPL network clause is triggered by a closed-source hosted service that uses ultralytics server-side, and the existence/cost/terms of the Ultralytics Enterprise license.
- google-genai SDK + the Google Gemini API (model gemini-2.5-flash). Distinguish the SDK license (Apache?) from the SERVICE terms. Confirm: (a) is commercial use of outputs allowed in a paid product; (b) does Google train on / human-review prompt+video data on the PAID/billed tier vs the free tier; (c) abuse-monitoring / data-retention terms; (d) current per-token / per-video pricing for gemini-2.5-flash (input video tokens-per-second, output) so we can sketch per-video cost.
- ffmpeg / ffprobe (system binaries, invoked as separate processes). Clarify GPL vs LGPL builds and whether pure server-side SaaS use (no distribution to users) triggers any source-disclosure obligation. SEPARATELY and importantly: video CODEC PATENT royalties for a commercial service that DECODES user-uploaded H.264/AVC and H.265/HEVC (e.g. GoPro footage) and ENCODES output ? covering MPEG-LA / Via-LA (AVC), Access Advance + Via-LA + others (HEVC) pools, AAC audio (Via-LA), what activities trigger royalties, any free thresholds (e.g. the AVC sub-100k or free-internet-broadcast provisions), and whether a small SaaS realistically incurs these. Then royalty-FREE codec alternatives for the OUTPUT we generate: AV1, VP9, Opus ? and their encoder maturity/speed on a GPU VM.
- torch, torchvision ? confirm BSD; flag that some torchvision \*pretrained weights\* may carry separate licenses.
- opencv-python ? confirm Apache-2.0 and that the PyPI wheel's bundled components are clean for commercial server use.
- lapx (a fork of "lap" for ByteTrack association) ? confirm license.
- Quick permissive confirmation for: fastapi, uvicorn, pydantic, jinja2, python-dotenv, numpy.

PERMISSIVE ALTERNATIVES TO RESEARCH (all must be GPU-VM-friendly and usable in a closed-source

commercial SaaS):

1. Object detection to replace ultralytics/YOLOv8 for player/person detection (and possibly shuttle): compare YOLOX (Apache-2.0), RT-DETR / RT-DETRv2 (the Baidu/Apache implementations ? NOT the Ultralytics port), PP-YOLOE, MMDetection (Apache-2.0), Detectron2 (Apache-2.0), torchvision built-in detectors (Faster R-CNN/RetinaNet/FCOS/SSD, BSD), RF-DETR (Roboflow), and D-FINE/DEIM. For EACH, verify BOTH the code license AND the pretrained-weights license (some "open" detectors like YOLO-NAS have restrictive weight licenses ? confirm). Note accuracy/speed trade-offs for person detection on sports video.
2. Shuttle/ball trajectory tracking ? VERIFY the actual repo licenses (commercial-use viability) of: WASB / WASB-SBDT ("Widely Applicable Strong Baseline" for sports ball detection), TrackNetV2 and TrackNetV3, MonoTrack, and any permissively-licensed alternative for tiny-fast-object tracking. This is the planned shuttle-tracking substrate, so its commercial-use license is high-stakes.
3. LLM / semantic rally-boundary step ? research ALL THREE strategies (the user wants all three compared):
  - (a) ELIMINATE the hosted LLM entirely: permissively-licensed temporal action segmentation / video models suitable for learning rally boundaries offline (e.g. MS-TCN/MS-TCN++, ASFormer, ActionFormer, and similar) ? verify licenses and whether they fit a from-annotations learned segmenter.
  - (b) SELF-HOST an open-weights vision-language model that can ingest video frames for semantic boundaries: compare Qwen2.5-VL, InternVL2/2.5, LLaVA-OneVision/LLaVA-Video, MiniCPM-V, VideoLLaMA ? for EACH verify the WEIGHTS license for commercial use (Apache-2.0 vs custom "research-only"/"non-commercial"/community licenses with MAU caps) and the GPU VRAM needed per size tier.
  - (c) KEEP Gemini but de-risk: summarize commercial terms, data-usage on paid tier, rate limits, and per-video cost at scale, plus other hosted-API alternatives (OpenAI, Anthropic) and how their commercial/data terms compare.

DELIVERABLE: A cited report containing (1) a per-dependency RISK TABLE ? columns: dependency, license, does-SaaS-hosting-trigger-obligations?, monetization verdict (CLEAR / WATCH / BLOCKER), recommended action; (2) a recommended FULLY-PERMISSIVE target stack that eliminates all BLOCKERS; (3) paid-license fallbacks where relevant; (4) a rough per-video processing-cost sketch on a GPU VM for each of the three LLM strategies; (5) an explicit list of the codec/patent considerations and the royalty-free output-codec recommendation. Prioritize accuracy and primary-source citations over breadth."

Your angle: **ffmpeg GPL/LGPL + video codec patent royalties (AVC/HEVC/AAC) and royalty-free outputs** ? Codec patent pools are a distinct legal/economic axis from software licensing; this angle establishes whether server-side decode/encode of user H.264/H.265 incurs Via-LA/Access Advance royalties, any free thresholds, and the royalty-free AV1/VP9/Opus output recommendation. Search query: `FFmpeg LGPL vs GPL build server-side SaaS no distribution; H.264 AVC MPEG-LA Via-LA and HEVC Access Advance Via-LA AAC patent royalties commercial decode encode service free thresholds; AV1 VP9 Opus royalty-free encoder GPU`

## Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

## 2. assistant (2026-05-30T03:03:26.288Z)

I'll research the ffmpeg licensing and codec patent royalty angle. Let me start by loading the WebSearch tool.

## 3. user (2026-05-30T03:03:47.012Z)

Web search results for query: "FFmpeg LGPL vs GPL build difference server-side use no distribution source disclosure"

Links: [{"title": "FFmpeg License and Legal

```
Considerations", "url": "https://www.ffmpeg.org/legal.html"}, {"title": "What's the difference between ffmpeg gpl version and lgpl version? · Issue #29 · BtbN/FFmpeg-Builds", "url": "https://github.com/BtbN/FFmpeg-Builds/issues/29"}, {"title": "Differences between GPL and LGPL when using licensed software - Fluendo", "url": "https://fluendo.com/blog/gpl-vs-lgpl/"}, {"title": "FFmpeg is licensed under LGPL which can (albeit carefully) work on the web, but ... | Hacker News", "url": "https://news.ycombinator.com/item?id=28252317"}, {"title": "Some parts of FFmpeg are LGPL, and some codecs are GPL (e.g., if you have x264 l... | Hacker News", "url": "https://news.ycombinator.com/item?id=27419783"}]
```

Based on the search results, here's what I found about FFmpeg LGPL vs GPL build differences:

## Build Configuration Differences

Compiling FFmpeg without "--enable-gpl" and without "--enable-nonfree" results in an LGPL build. The GPL build includes additional codecs and libraries, notably GPL libraries like libx264.

## Linking and Distribution Requirements

The key difference in requirements depends on your linking method:

- **\*\*LGPL build with dynamic linking\*\***: As long as the application is linked dynamically to LGPL software, the proprietary code can be kept proprietary. However, you must distribute the source code of FFmpeg, no matter if you modified it or not.
- **\*\*LGPL build with static linking\*\***: If linked statically, the application code must also be released as LGPL, or everything that allows the user to re-link the application with a different version of the LGPL source code must be provided.

## Server-Side Use Considerations

Importantly, services that merely use ffmpeg within their own organization are not under GPL/LGPL source code distribution obligations, even if providing a service to third parties. This means server-side use without distributing the binary to end users is treated differently from distributing compiled binaries.

The LGPL vs GPL difference primarily determines which additional codecs are available in your build, and the licensing obligations depend heavily on whether you dynamically or statically link and whether you distribute the compiled software.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 4. user (2026-05-30T03:04:17.034Z)

Web search results for query: "H.264 AVC patent license Via-LA MPEG-LA royalty commercial encode decode service free threshold 100000 units"

Links: [{"title": "As a small business, must I pay royalties for H264 and H265? ? Streaming For The Poor", "url": "https://streaming4thepoor.live/as-a-small-business-must-i-pay-royalties-for-h264-and-h265/"}, {"title": "FOSS Patents: MPEG LA's AVC/H.264 licensing terms: further analysis", "url": "http://www.fo SSPatents.com/2010/06/mpeg-las-avch264-licensing-terms.html"}, {"title": "Advanced Video Coding - Wikipedia", "url": "https://en.wikipedia.org/wiki/Advanced\_Video\_Coding"}, {"title": "What Is H.264?", "url": "https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=74735"}, {"title": "Settle your questions about H.264 license cost once and for all (hopefully) | by Jina Jiayang Liu | Medium", "url": "https://jina-liu.medium.com/settle-your-questions-about-h-264-license-cost-once-and-for-all-hopefully-a058c2149256"}, {"title": "AVC/H.264 - Via Licensing Alliance", "url": "https://www.via-la.com/licensing-programs/avc-h-264/"}, {"title": "MPEG LA - Wikipedia", "url": "https://en.wikipedia.org/wiki/MPEG\_LA"}, {"title": "Enhanced Mode Selection Algorithm for H.264 encoder for Application in Low Computational power devices", "url": "https://arxiv.org/pdf/0909.0245"}]

Based on the search results, here's the key information about H.264/AVC patent licensing:

## Patent Administration

A private organization known as Via-LA administers the licenses for patents applying to the H.264/AVC standard, as well as other patent pools such as for MPEG-4 Part 2 Video, HEVC and MPEG-DASH.

## Royalty Thresholds and Free Use Cases

For products sold or distributed with AVC encoders or decoders, royalties start after the first 100,000 units are sold each year, and cap at \$3.5 million per year 2005-2006, \$4.25 million per year 2007-08, \$5 million per year 2009-10, and \$6.5 million per year in 2011-15.

For subscription services, royalties scale with the number of subscribers, but only start after exceeding 100,000 subscribers.

On August 26, 2010, MPEG-LA announced that royalties would not be charged for H.264 encoded Internet video that is free to end users.

## Licensing Structure

The License employs annual limitations to provide cost predictability, threshold levels below which certain royalties will not be charged in order to encourage early-stage adopters and minimize the impact on lower volume users or demo products.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 5. user (2026-05-30T03:04:18.839Z)

Web search results for query: "HEVC H.265 patent royalties Access Advance Via-LA decode encode service commercial 2024 free threshold"

Links: [{"title": "HEVC Advance - Access Advance", "url": "https://accessadvance.com/licensing-programs/hevc-advance/"}, {"title": "HEVC/VVC - Via Licensing Alliance", "url": "https://www.via-la.com/licensing-programs/hevc-vvc/"}, {"title": "Decoding the Landscape: Recent Developments in Video Codec Licensing", "url": "https://www.streamingmediaglobal.com/Articles/Editorial/Featured-Articles/Decoding-the-Landscape-Recent-Developments-in-Video-Codec-Licensing-164043.aspx"}, {"title": "HEVC Advance Patent Pool: - Access Advance", "url": "https://accessadvance.com/hevc-advance-patent-pool-detailed-royalty-rates/"}, {"title": "Access Advance Extends HEVC Advance Rate Increase Deadline", "url": "https://www.businesswire.com/news/home/20260126799824/en/Access-Advance-Extends-HEVC-Advance-Rate-Increase-Deadline"}, {"title": "Home - Access Advance", "url": "https://accessadvance.com/"}, {"title": "Access Advance Announces HEVC Advance and VVC Advance Pricing through 2030 - Access Advance", "url": "https://accessadvance.com/2025/07/21/access-advance-announces-hevc-advance-and-vvc-advance-pricing-through-2030/"}, {"title": "To Deploy HEVC, Users Must Choose What Patent Pool To Dive Into - The Broadcast Bridge - Connecting IT to Broadcast", "url": "https://www.thebroadcastbridge.com/content/entry/11204/to-deploy-hevc-users-must-choose-what-patent-pool-to-dive-into"}, {"title": "Access Advance Addresses Düsseldorf Court Ruling, Announces Revised Duplicate Royalty Policy for the HEVC Advance Patent Pool", "url": "https://www.businesswire.com/news/home/20220308006185/en/Access-Advance-Addresses-Düsseldorf-Court-Ruling-Announces-Revised-Duplicate-Royalty-Policy-for-the-HEVC-Advance-Patent-Pool"}]

Based on the search results, here's what I found regarding HEVC H.265 patent royalties with Access Advance, Via-LA, and related information:

## Patent Pool Administration

Access Advance has acquired the administrator of Via LA's HEVC/VVC patent pool, now named the

VCL Advance (Video Codec Licensing) program. Access Advance manages and administers the HEVC Advance Patent Pool for licensing over 29,000 patents essential to HEVC/H.265 technology.

## Content Royalties and Free Content Threshold

An important distinction exists regarding content licensing: The Sisvel, VIA-LA, and Access Advance pools don't charge for content as a strict decision of the patent owners, who call the shots. However, back in 2015, patent owners in the HEVC pool thought it best to forgo royalties on content to help drive adoption.

## Encoding and Decoding Services

The VDP Pool is focused on licensing video streaming, i.e., the use of the codecs to encode or transmit commercial video content in video streaming services. This separate pool covers both encode and decode operations for streaming services.

## Recent Rate Changes

Companies that become Licensees of the HEVC Advance program on or before June 30, 2026 will secure current royalty rates. This is a temporary postponement of the 25% increase in rates and caps applicable to Licensees who sign up after the previous December 31, 2025 deadline.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 6. user (2026-05-30T03:04:38.307Z)

Web search results for query: "AV1 VP9 Opus royalty-free codec NVIDIA GPU NVENC AV1 encoder hardware support maturity"

Links: [{"title":"What Is The AV1 Codec, Which Graphics Cards Support It And Why It Matters | HotHardware","url":"https://hothardware.com/news/av1-codec-support-and-importance-explained"}, {"title":"Improving Video Quality and Performance with AV1 and NVIDIA Ada Lovelace Architecture | NVIDIA Technical Blog","url":"https://developer.nvidia.com/blog/improving-video-quality-and-performance-with-av1-and-nvidia-ada-lovelace-architecture/"}, {"title":"Video Codecs Explained: H.264, H.265, AV1 & VP9","url":"https://antmedia.io/video-codecs-streaming-guide/"}, {"title":"Blackmagic Forum ? View topic - AV1 Encoding in Resolve 19 (Free)","url":"https://forum.blackmagicdesign.com/viewtopic.php?f=21&t=207091"}, {"title":"Video Codec SDK | NVIDIA Developer","url":"https://developer.nvidia.com/video-codec-sdk"}, {"title":"AV1 Codec Explained: What Is It and How It's Used","url":"https://www.videoproc.com/resource/av1-codec.htm"}, {"title":"HWAAccelIntro ? FFmpeg","url":"https://trac.ffmpeg.org/wiki/HWAAccelIntro"}, {"title":"Encode/AV1 ? FFmpeg","url":"https://trac.ffmpeg.org/wiki/Encode/AV1"}]

Based on the search results, here's information about royalty-free codecs and NVIDIA GPU hardware support:

## Royalty-Free Codecs

AV1 is royalty-free and open source, coming from the Alliance for Open Media (AOMedia) which formed in 2015 for the explicit purpose of creating an open alternative. VP9 has gained reasonable adoption including hardware encoding support on many platforms, but ultimately is regarded as often less efficient than H.265.

## AV1 Codec Advantages

AV1 achieves 30-50% better compression than H.265 at equivalent quality, with research demonstrating AV1 encoding at 1,500 kbps matching H.265 quality at 2,250 kbps for 1080p content. NVIDIA NVENC AV1 encoding results in a 40% bit rate savings over H.264 at 1080p60.

## NVIDIA GPU Hardware Support

NVIDIA Ampere architecture introduced hardware-accelerated AV1 decoding, while NVIDIA Ada Lovelace architecture supports both AV1 encoding and decoding. NVDEC supports hardware-accelerated decoding of the following video codecs on Windows and Linux platforms: MPEG-2, VC-1, H.264 (AVCHD), H.265 (HEVC), VP8, VP9, and AV1.

## Hardware Support Maturity

Hardware support remains limited compared to H.264. However, the software enhancements in Video Codec SDK v13.0 enable extending ultra-high quality (UHQ) mode to AV1 encoding, making AV1 encoding on NVIDIA Blackwell NVENC comparable to software AV1 encoding with ~3X throughput.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 7. user (2026-05-30T03:05:39.975Z)

Web search results for query: "SVT-AV1 encoder Apache license speed AV1 software encoding server CPU vs NVENC quality"

Links: [{"title": "Download SVT-AV1 Encoder Free - Fast AV1 Video Encoding Software", "url": "https://www.free-codecs.com/download/svt-av1.htm"}, {"title": "The SVT-AV1 encoder: overview, features and speed-quality tradeoffs - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2020SPIE11510E..21K/abstract"}, {"title": "SVT-AV1: open-source AV1 encoder and decoder | by Netflix Technology Blog | Netflix TechBlog", "url": "https://netflixtechblog.com/svt-av1-an-open-source-av1-encoder-and-decoder-ad295d9b5ca2"}, {"title": "Encode/AV1 ? FFmpeg", "url": "https://trac.ffmpeg.org/wiki/Encode/AV1"}, {"title": "SVT-AV1 3.0 Released With Faster CPU-Based AV1 Encoding - Phoronix", "url": "https://www.phoronix.com/news/SVT-AV1-3.0-Released"}, {"title": "SVT-AV1 Encoding Guide · GitHub", "url": "https://gist.github.com/dvaupel/716598fc9e7c2d436b54ae00f7a34b95"}, {"title": "SVT-AV1 | Codec Wiki", "url": "https://wiki.x266.mov/docs/encoders/SVT-AV1"}, {"title": "SVT-AV1 4.1 Download Free - VideoHelp", "url": "https://www.videohelp.com/software/SVT-AV1"}, {"title": "Encoder Complexity Control in SVT-AV1 by Speed-Adaptive Preset Switching", "url": "https://arxiv.org/pdf/2307.05208"}]

Based on the search results, here's what I found about SVT-AV1:

## License

SVT-AV1 is an open-source AV1 codec implementation under a BSD + patent license, not Apache license as mentioned in your query.

## Encoding Speed

SVT-AV1 is an open-source software AV1 encoder architected to yield excellent quality-speed-latency tradeoffs on CPU platforms for a wide range of video coding applications. SVT-AV1 leverages multi-threading and hardware acceleration to encode videos up to 10x faster than traditional AV1 encoders.

## Development & Leadership

SVT-AV1 is the open-source, CPU-based AV1 encoder that was started by Intel software engineers and now led by the Alliance for Open Media.

## Quality & Performance

SVT-AV1 reaches similar compression efficiency as libaom at the slowest speed settings. The encoder supports a wide range of speed-efficiency tradeoffs and scales fairly well across many CPU cores.

## CPU vs GPU Encoding

The search results indicate that SVT-AV1 is specifically a CPU-based software encoder. While one result mentions "NVEnc AV1 - NVIDIA GPU-accelerated encoding" as an alternative, the results don't provide detailed comparisons between SVT-AV1's CPU performance and NVIDIA's NVENC GPU encoding in terms of quality.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

**8. user (2026-05-30T03:06:05.607Z)**

Structured output provided successfully

**9. user (2026-05-30T03:07:40.090Z)**

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

**10. user (2026-05-30T03:07:41.218Z)**

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

**11. user (2026-05-30T03:07:41.913Z)**

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

**12. user (2026-05-30T03:07:42.376Z)**

## FFmpeg Licensing Summary

### LGPL vs GPL

FFmpeg is primarily under LGPL v2.1, but contains optional GPL v2 components. The key statement:

"If those parts get used the GPL applies to all of FFmpeg."

### Server-Side Use & Source Disclosure

The compliance checklist focuses on distribution scenarios. There is no explicit mention of server-side use exceptions. The document states you must "Distribute the source code of FFmpeg, no matter if you modified it or not" when distributing binaries?implying obligations attach to distribution, not mere use.

### Patent & Codec Disclaimers

The page includes a candid patent disclaimer:

"We do not know, we are not lawyers so we are not qualified to answer this."

Regarding commercial use with potentially patented codecs:

"Once you start trying to make money from patented technologies, the owners of the patents will come after licensing fees."

The document notes that standards like H.264 and MPEG-4 contain patent warnings, but FFmpeg itself takes no position on patent validity or applicability across jurisdictions.

**13. user (2026-05-30T03:07:45.689Z)**

## H.264/AVC Patent Licensing Summary

**\*\*Administrator:\*\*** Via Licensing Authority (formerly MPEG LA) administers the patent pool. The article notes that "Via Licensing Corp acquired MPEG LA in April 2023" and established a new administration company called Via Licensing Alliance.

**\*\*Free Internet Video Provision:\*\*** On August 26, 2010, MPEG-LA announced that "royalties would not be charged for H.264 encoded Internet video that is free to end users." However, this exemption has important limits.

**\*\*Royalties Required:\*\*** The document specifies that "All other royalties remain in place, such as royalties for products that decode and encode H.264 video as well as to operators of free television and subscription channels." This means encoding, decoding, and distributing video through paid or broadcast channels all trigger licensing fees.

**\*\*Cisco's Approach:\*\*** As a notable exception, "Cisco paid royalties to MPEG LA on behalf of the users of binaries" for its open-source OpenH264 encoder, making those binaries royalty-free to users.

**\*\*Patent Holders:\*\*** Major patent holders include Panasonic (1,197 patents), Godo Kaisha IP Bridge (1,130 patents), and LG Electronics (990 patents).

**\*\*Future Terms:\*\*** The article mentions that "In February 2026, the VIA Licensing Authority announced a new licensing scheme targeting video streaming" with updated terms for new licensees after January 2026.

14. user (2026-05-30T03:07:46.673Z)

## HEVC Patent Licensing Summary

### Patent Pools

**\*\*MPEG LA\*\*** (announced September 29, 2014):

- Covers essential patents from 23 companies
- Device royalty: First 100,000 units royalty-free; thereafter \$0.20 per device up to \$25 million annually
- No content royalties

**\*\*HEVC Advance\*\*** (announced March 26, 2015):

- Initially covered 500 essential patents from AT&T, Microsoft, Nokia, and Motorola
- Original rates: \$2.60/device (Region 1), \$1.30/device (Region 2), plus 0.5% content revenue royalty
- December 2015 revision: Reduced to \$2.03/device (Region 1) with annual caps (\$40M devices, \$5M content, \$2M optional features)
- November 2022: Allowed royalty-free software distribution to consumer devices

**\*\*Velos Media\*\*** (announced March 31, 2017):

- Covers essential patents from Ericsson, Panasonic, Qualcomm, Sharp, and Sony

### Cost Comparison

Combined MPEG LA and HEVC Advance licensing totaled approximately "\$2.80" per device?described as "twenty-eight times as expensive as AVC." This fragmentation contrasted sharply with the unified approach, prompting major tech companies to pursue the royalty-free **\*\*AV1\*\*** alternative, which was finalized March 28, 2018.

15. user (2026-05-30T03:07:49.588Z)

## AVC/H.264 Royalty Terms for Encoding/Decoding Services

### Codec Manufacture and Sales (Section 1)



For products sold to end users and OEMs for PCs:

**\*\*Volume Thresholds and Per-Unit Fees:\*\***

- "For the first 1 to 100,000 units: \$0.00" (free threshold)
- Units 100,001 to 5,000,000: \$0.20 each
- Units 5,000,001+: \$0.10 each

**\*\*Enterprise Annual Caps:\*\***

- 2017 and after: "\$9.75M per year"

## Streaming (Section 2)

**\*\*Title-by-Title Streaming:\*\***

- Content 12 minutes or less: Free
- Longer content: Lower of (a) 2% of remuneration or (b) \$0.02 per title

**\*\*Tiered Streaming Licenses\*\*** (applies to OTT, FAST, social media, cloud gaming, cable/satellite, broadcast):

- Tier 1 (largest): "\$4,500,000" annually
- Tier 2: "\$3,375,000" annually
- Tier 3: "\$2,250,000" annually
- Small/nascent services: "\$100K" annually

**\*\*Enterprise Cap (streaming):\*\*** "\$9.75M per year in 2017 and after"

### 16. assistant (2026-05-30T03:08:00.699Z)

I have already completed the research and submitted my findings via StructuredOutput. The task is complete.

To summarize what the research established on the ffmpeg + codec-patent angle:

- **\*\*FFmpeg licensing\*\***: Core is LGPL-2.1; `--enable-gpl` (x264/x265) makes it GPL; `--enable-nonfree` makes it non-redistributable. FFmpeg uses GPL/LGPL, NOT AGPL ? so a pure server-side SaaS that invokes ffmpeg as a separate process and never distributes the binary triggers no source-disclosure obligation. FFmpeg's own legal page disclaims patent licenses entirely.
- **\*\*AVC/H.264 (Via LA)\*\***: First 100,000 units/yr free, ~\$0.20/unit, ~\$9.75M annual cap; free-to-end-user internet video carries no royalty, but products/operators that encode/decode still nominally trigger fees. Small SaaS realistically falls under thresholds.
- **\*\*HEVC/H.265\*\***: Fragmented across Access Advance, Via LA, and Velos Media plus independents ? ~\$2.80/device combined ("28x AVC"). This is the highest-uncertainty patent item (decoding user GoPro HEVC).
- **\*\*Royalty-free outputs\*\***: AV1 (SVT-AV1, BSD-3-Clear + AOM patent license), VP9 (libvpx, BSD), Opus (BSD, IETF royalty-free). AV1 hardware NVENC encode needs Ada-class GPUs (L4/L40); T4/A100 can decode AV1 but must software-encode via SVT-AV1.

The findings (6 ranked, cited results) were already returned through the StructuredOutput tool.